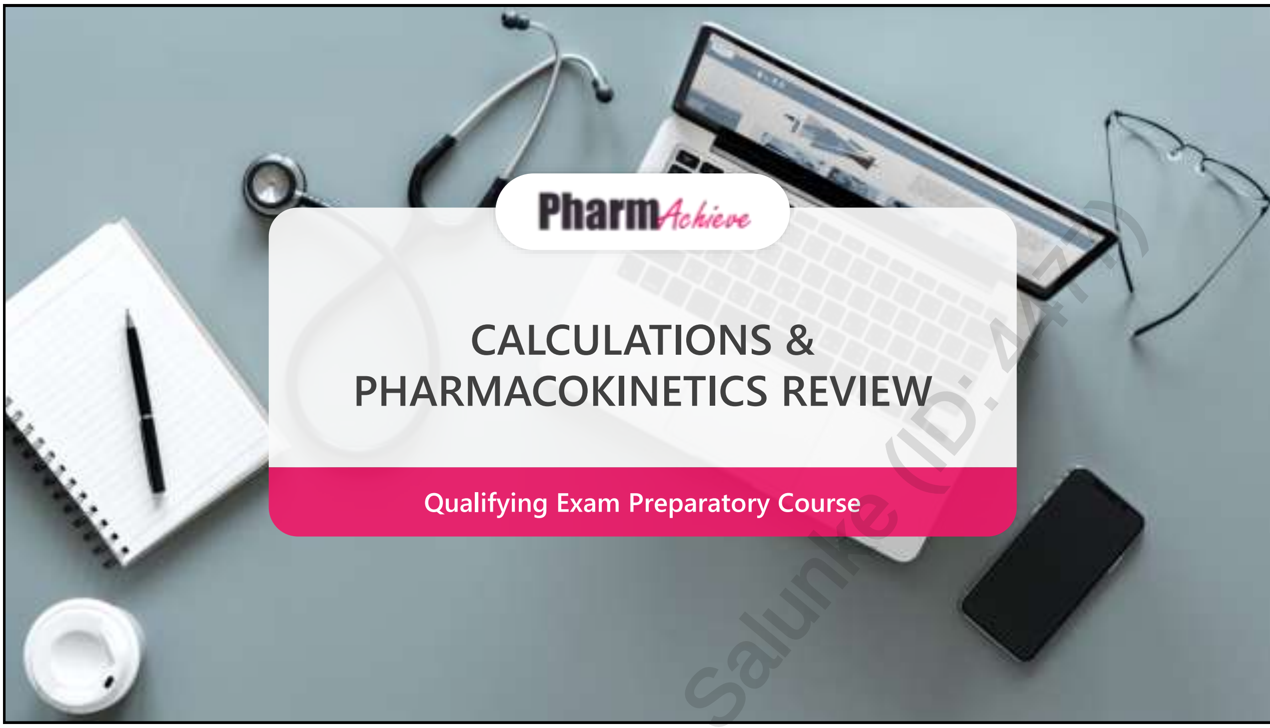


1

COVERED IN THIS PRESENTATION…

- Ethical, Legal and Professional Responsibilities
- Patient Care
- **Product Distribution**
- Practice Setting
- Health Promotion
- Knowledge and Research Application
- Communication and Education
- Intra and Inter-Professional Collaboration
- Quality and Safety

Source: pebc.ca

2

LEGEND

ATP: Adenosine Triphosphate
AUC: Area Under The Curve
CL: Clearance
Cp: Plasma Concentration
D: Dose
F: Bioavailability
IBW: Ideal Body Weight
IM: Intramuscular
IV: Intravenous
kg: Kilogram
L: Liter
lb: Pound
PK: Pharmacokinetics
PO: *Per os* (oral)
SC: Subcutaneous
TDM: Therapeutic Drug Monitoring
Vd: Volume of Distribution
w/v: Weight/Volume
w/w: Weight/Weight
τ: Dosing Interval

3

EXPONENTS AND LOGARITHMS

Form = X^a
X is the base
A is the exponent
$If\ X^b = Y, then\ \log_X Y = b$

- $X^0 = 1, X^1 = X$
- $X^{-a} = \frac{1}{X^a}$
- $X^a * X^b = X^{a+b}$
- $\frac{X^a}{X^b} = X^{a-b}$
- $X^{a^b} = X^{a*b}$
- $(X \cdot Y)^a = X^a \cdot Y^a$

- $\log_a a^x = x$
- $\log(XY) = \log X + \log Y$
- $\log \frac{X}{Y} = \log X - \log Y$
- $\log(X^y) = y \log X$
- $\log_a X = \frac{\log_b X}{\log_b a}$
- $\log(X + Y) \neq \log X + \log Y$

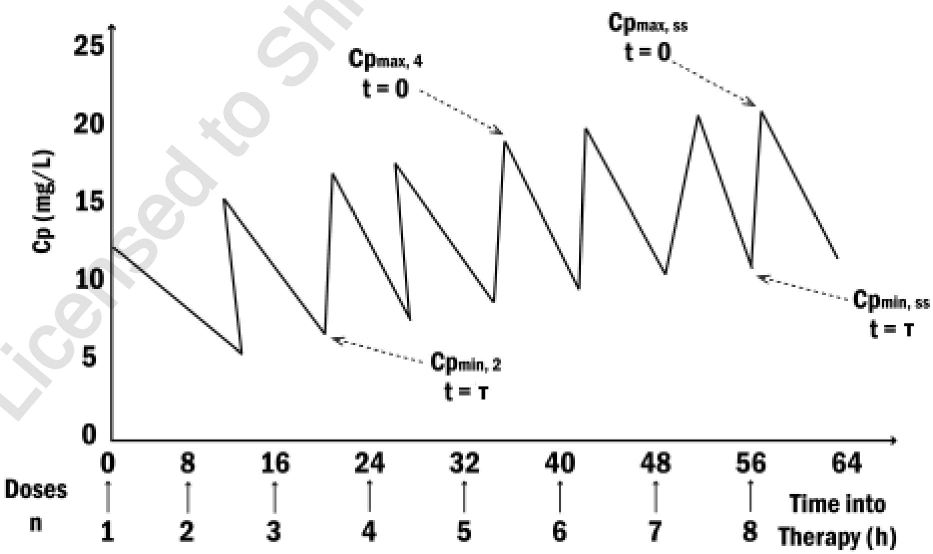
4

IV BOLUS, MULTIPLE DOSING

Qualifying Exam Preparatory Course

5

PLASMA CONCENTRATION-TIME PROFILE



6

STEADY-STATE BASIC EQUATION

The steady-state basic equation is:

$$Cp_{ss,n} = \frac{F \cdot D}{Vd \cdot (1 - e^{-k\tau})} \cdot e^{-kt}$$

- $Cp_{ss,n}$ is the plasma concentration
- k is the elimination rate constant
- τ is the dosing interval
- F is the bioavailability factor
- Vd is the volume of distribution
- D is the dose
- t is the time

7

STEADY-STATE BASIC EQUATION

The steady-state basic equation for the maximum plasma concentration:

$$Cp_{max,ss} = \frac{F \cdot D}{Vd \cdot (1 - e^{-k\tau})}$$

- $Cp_{max,ss}$ is the maximum plasma concentration at steady state
- k is the elimination rate constant
- τ is the dosing interval
- F is the bioavailability factor
- Vd is the volume of distribution
- D is the dose

8

STEADY-STATE BASIC EQUATION

The steady-state basic equation for the minimum plasma concentration:

$$Cp_{min,ss} = Cp_{max,ss} \cdot e^{-k\tau}$$

- $Cp_{max,ss}$ is the maximum plasma concentration at steady state
- k is the elimination rate constant
- τ is the dosing interval
- $Cp_{min,ss}$ is the minimum steady-state plasma concentration

9

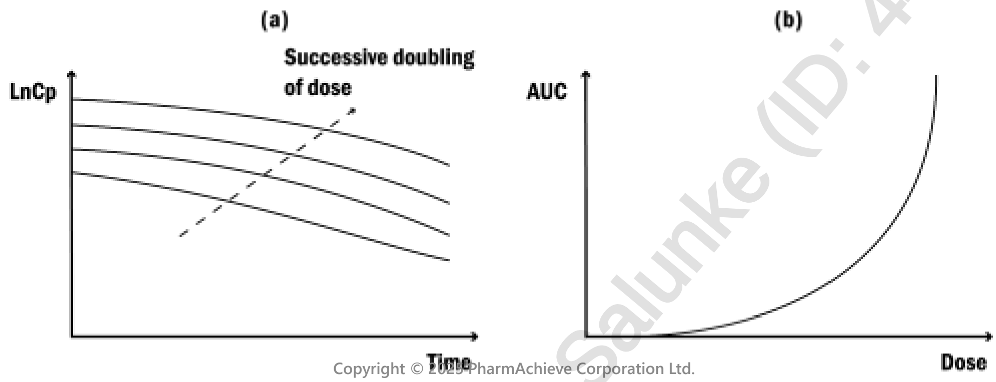
NON-LINEAR KINETIC DRUGS

Linear kinetics	Non-linear kinetics
Rate of elimination proportional to C_p and Dose	Constant rate of elimination, irrespective of C_p or dose
Normal behavior for most drugs	Caused by saturation of transporters or enzymes
Changes in dosage parallels change in C_p and AUC; 2x increase in Dose = 2x increase in C_p and AUC	Change in dosage may have much bigger changes in C_p and AUC; 2x increase in dose > 2x increase in C_p and AUC
Variations in dosage is relatively OK if within the therapeutic window	Small changes in dosage can cause toxicity or loss of efficacy
Requires no special monitoring	Blood C_p should be monitored and renal correction may be used (Therapeutic Drug Monitoring)

10

NON-LINEAR KINETIC DRUGS

Impact of non-linear elimination on the plasma concentration–time profile after single doses. Graph (a) is a semi-logarithmic plot of plasma concentration against time after a successive doubling of an intravenous dose. After large doses, the initial fall in the plasma concentration with time is less steep than with linear pharmacokinetics. Graph (b) shows that the area under the plasma concentration–time curve (AUC) increases with dose for drugs that display saturable elimination.



SATURABLE KINETIC DRUGS

Drugs that are subject to saturable kinetics have a high risk of displaying non-linear kinetics

Drug	Uses and effect	Toxicity
Theophylline	Treatment of COPD Relaxes smooth muscle, ↑ respiration & HR	Nausea, vomiting, ionic imbalance Metabolic acidosis & tachycardia
Digoxin	Na+/K+ ATPase inhibitor Increases cardiac contractility Treatment of arrhythmia	Tachycardia or fibrillation Nausea and vomiting Ocular disturbance
Procainamide	Rapid block of sodium channels Class IA antiarrhythmic	Immune system failure Pacemaker failure
Phenytoin	Sodium channel blocker Treatment of seizures and heart arrhythmias	Low BP Abnormal heart rhythms
Salicylic Acid	Active form of aspirin Commonly used NSAID	Primary metabolic acidosis Primary respiratory alkalosis Nausea, tachycardia, hyperthermia

PHENYTOIN

- Phenytoin is commonly prescribed and displays nonlinear PK
- Has a narrow therapeutic range and requires individualized dosing regimens
- **Dosing should be calculated based on IBW**
- Phenytoin is primarily eliminated by saturable metabolism → described by Michaelis-Menten equation
- Average Km: 4 mg/mL
- Average V_{max}: 7mg/kg/day

13

ORAL ADMINISTRATION OF PHENYTOIN

The following equation can be used for phenytoin and other drugs that have similar saturable elimination:

$$F \cdot D = \frac{V_{max} \cdot Cp_{SS}}{Km + Cp_{ss}}$$

- V_{max} is the maximum metabolism speed achieved at enzyme saturation
- Km is the Michaelis-Menten constant
- Cp_{ss} is the steady-state plasma concentration
- F is the bioavailability
- D is the dose

14

PHENYTOIN STEADY STATE CONCENTRATION

The steady-state plasma concentration can be found by rearranging the previous formula:

$$Cp_{ss} = \frac{Km \cdot F \cdot R_a}{V_{max} - F \cdot R_a}$$

- V_{max} is the maximum metabolism speed achieved at enzyme saturation
- Km is the Michaelis-Menten constant
- Cp_{ss} is the steady-state plasma concentration
- F is the bioavailability
- R_a is the rate of drug administration

15

PHENYTOIN MAXIMUM SPEED

The maximum speed of metabolism can be found by rearranging the previous formula:

$$V_{max} = \frac{(Km + Cp_{ss}) \cdot F \cdot R_a}{Cp_{ss}}$$

- V_{max} is the maximum metabolism speed achieved at enzyme saturation
- Km is the Michaelis-Menten constant
- Cp_{ss} is the steady-state plasma concentration
- F is the bioavailability
- R_a is the rate of drug administration

16

PRACTICE PROBLEM 10

Determine an intravenous loading dose of phenytoin sodium injection for AB who is 5’11”, 71 kg, using a Vd of 0.65 L/kg and a desired level of 20 mg/L.

17

PRACTICE PROBLEM 10A

Determine an intravenous loading dose of phenytoin sodium injection for AB (male patient) who is 5’11”, 71 kg, using a Vd of 0.65 L/kg and a desired level of 20 mg/L.

Step 1: Calculate IBW

IBW $= \mathbf{50\ kg + 2.3kg\ * (\#\ of\ inches\ over\ 5\ ft)}$

IBW $= \mathbf{50\ kg + 2.3kg\ * (11)}$

IBW = 75.3kg (patient’s actual body weight < ideal body weight, so use actual body weight)

Step 2: Calculate patient specific Vd

Vd $= \mathbf{0.65 \frac{L}{kg} (71\ kg)}$

Vd $= \mathbf{46.15\ L}$

Step 3: Calculate dose

Dose $= \frac{C * Vd}{S * F} = \frac{\frac{20mg}{L} * 46.15\ L}{0.92 * 1}$

Dose = 1003 mg

18

PRACTICE PROBLEM 10B

Determine a daily maintenance dose of oral phenytoin sodium capsules for AB using a Vmax of 10.18 mg/kg/day, Km of 10.9 mg/L, Css = 12.3 mg/L. Round to the nearest dose using available phenytoin sodium capsules (100 mg extended-release capsules) to determine a daily dose for AB. Assume full bioavailability (F=1).

19

PRACTICE PROBLEM 10B

Determine a daily maintenance dose of oral phenytoin sodium capsules for AB using a Vmax of 10.18 mg/kg/day, Km of 10.9 mg/L, Css = 12.3 mg/L. Round to the nearest dose using available phenytoin sodium capsules (100 mg extended-release capsules) to determine a daily dose for AB. Assume full bioavailability (F=1).

Step 1: Calculate patient specific Vmax

71 kg * 10.18 mg/kg/day = 722.78 mg/day

Step 2: Plug in givens

$$F \cdot D = \frac{V_{max} \cdot Cp_{SS}}{Km + Cp_{ss}}$$

$$D = \frac{722.78 \cdot 12.3\,\frac{mg}{L}}{10.9\,\frac{mg}{L} + 12.3\,\frac{mg}{L}} = 383\,\frac{\text{mg}}{day} \sim 400\,\frac{mg}{day}$$

20

PRACTICE, PRACTICE, PRACTICE

Qualifying Exam Preparatory Course

21

Ask An Instructor – Question (PK Quiz #10 – ID 3813)

If a drug has a therapeutic window of 150 mg/L and a MTC (minimum toxic concentration) of 500 mg/L, what is the dosage interval for IV administration to maintain an effective Cp at all times given that the Kel is 0.9/h?

22

QUESTION #1 – OPIOID CONVERSION (i.e. IV to PO)

AT, a 45-year-old male, has been receiving hydromorphone IV 1.5 mg every 4 hours for severe post-operative pain. As his pain is improving, he is being prepared for discharge, and the physician wants to switch him to oral morphine for continued pain management at home. Options for oral morphine include extended-release formulation every 12 hours or immediate-release formulation every 4-6 hours.

23

DOSAGE CALCULATIONS

OPIOID CONVERSIONS

- Step 1: Ensure the formulation is consistent (PO vs. IV)
- Step 2: Use the chart to convert from one opioid to another
- Step 3: Reduce the new dose by 25-50% due to incomplete cross-tolerance (few exceptions)

Drug	Oral dose (mg)	IV/IM/SC dose (mg)
Morphine	30	10
Oxycodone	20	-
Hydromorphone	6	1.5
Codeine	200	-
Tramadol	150	-
Methadone	Morphine dose equivalence not reliably established	

24

QUESTION #1 – OPIOID CONVERSION (i.e. IV to PO)

I. What is the total daily dose of IV hydromorphone that AT is currently receiving?

a) 6 mg/day
b) 9 mg/day
c) 36 mg/day
d) 12 mg/day

II. What is the equivalent daily dose of oral morphine that AT would need to be taking?

a) 180 mg/day
b) 145 mg/day
c) 135 mg/day
d) 170 mg/day

25

QUESTION #1 – OPIOID CONVERSION (i.e. IV to PO)

III. What would be an appropriate oral morphine dose and regimen? You have 10, 15, 30, 60, 100 or 200 mg extended-release capsules available.

a) 75 mg qam + 60 mg qpm
b) 60 mg BID
c) 70 mg BID
d) 90 mg BID

26

QUESTION #2 – ALLIGATION, CROSS-SUBTRACT METHOD

You are working in a hospital pharmacy, and you receive an order to prepare 500 mL of a 25% dextrose (D25W) solution for an adult patient. However, the stock solutions available in your pharmacy are: dextrose 70%, dextrose 5%. You need to mix these two available solutions to create the required 25% dextrose solution.

27

QUESTION #2 – ALLIGATION, CROSS-SUBTRACT METHOD

How many milliliters of the D70W solution and how many milliliters of the D5W solution should you mix to make 500 mL of a 25% dextrose solution?

a) 346.15 mL
b) 243.85 mL
c) 153.85 mL
d) 256.15 mL

28

SOLVE USING CROSS-SUBTRACT METHOD (IN-CLASS)

29

QUESTION #3 – ALLIGATION, CROSS-SUBTRACT METHOD

You need to prepare 1,000 mL of a 30% sodium chloride (NaCl) solution.

Available stock solutions: 70% sodium chloride, 50% sodium chloride, 20% sodium chloride.

You need to determine how much of each stock solution should be mixed to create the required 1,000 mL of 30% sodium chloride.

30

QUESTION #3 – ALLIGATION, CROSS-SUBTRACT METHOD

You need to prepare 1,000 mL of a 30% sodium chloride (NaCl) solution.

Available stock solutions: 70% sodium chloride, 50% sodium chloride, 20% sodium chloride.

You need to determine how much of each stock solution should be mixed to create the required 1,000 mL of 30% sodium chloride.

70%		10 parts
50%		10 parts
	30%	
20%		20+40= 60 parts
		80 parts total

We need…
- 125 mL of 70% and 50% each
- 750 mL of 20% solution

31

SOLVE USING SYSTEMS OF EQUATION (IN-CLASS)

32

BIOAVAILABILITY (F)

- Fraction of the dose that reaches systemic circulation ("how much of the drug the body sees")
- Ranges between 0 (poor or negligible F) and 1 (100% available)

$$F = \frac{(AUC_{\infty}/Dose)}{(AUC_{\infty,iv}/Dose_{iv})}$$

- Two terms:
  - Absolute F: bioavailability calculated relative to an IV dose (which has F=1)
  - Relative F: bioavailability calculated relative to another route of administration

33

QUESTION #4 – RELATIVE BIOAVAILABILITY

A patient has been prescribed a new medication that can be administered either orally (PO) or by intramuscular injection (IM). To determine the oral bioavailability of the drug, a pharmacokinetic study was conducted, measuring the area under the curve (AUC) for both the oral and intramuscular formulations.

The study provided the following data:

- AUC (IM): 300 µg·h/mL after a 100 mg intramuscular injection
- AUC (PO): 180 µg·h/mL after a 150 mg oral dose

Calculate the oral bioavailability (F) in % of the drug using the AUC values from the study.

a) 30%
b) 40%
c) 50%
d) 60%

34

QUESTION #5

JM is a 60 kg patient who is started on an IV infusion of amikacin at a rate of 5 mg/min. Assume a volume of distribution of 30 L, a half–life of 1 h, and a clearance of 20 L/h. What would be the plasma concentration of amikacin at steady state?

35

QUESTION #5 - ANSWER

JM is a 60 kg patient who is started on an IV infusion of amikacin at a rate of 5 mg/min. Assume a volume of distribution of 30 L, a half–life of 1 h, and a clearance of 20 L/h. What would be the plasma concentration of amikacin at steady state?

Equation for IV infusion: $Cp = \frac{R_0}{Cl} \cdot (1 - e^{-kt})$

At steady state: $Cp = \frac{R_0}{Cl}$

5 mg/min x (60 min/1hour) = 300 mg/hr

$Cp = \frac{R_0}{Cl} = \frac{300\ mg/hr}{20\ L/hr}$ = 15 mg/L

36

QUESTION #6

MR is a 30-year-old female who weighs 55 kg and has been prescribed a single dose Drug Y at 20 mg/kg. Drug Y has an elimination half-life of 5 hours, a volume of distribution of 0.25 L/kg, and is 30% protein bound. Drug Y follows a one-compartment model and first-order kinetics.

- Calculate the initial plasma concentration (Cp_0) of Drug Y
- Predict the plasma concentration of Drug Y after 10 hours
- Calculate how much of the drug remains in the body after 10 hours

37

QUESTION #6 - ANSWER

MR is a 30-year-old female who weighs 55 kg and has been prescribed a single dose Drug Y at 20 mg/kg. Drug Y has an elimination half-life of 5 hours, a volume of distribution of 0.25 L/kg, and is 30% protein bound. Drug Y follows a one-compartment model and first-order kinetics.

Calculate the initial plasma concentration (Cp_0) of Drug Y
Dose of drug Y = 20 mg/kg * 55 kg = 1100 mg
Vd of MR = 0.25 L/kg * 55 kg = 13.75 L
Cp0 = 1100 mg / 13.75 L = 80 mg/L

Predict the plasma concentration of Drug Y after 10 hours
At t=0, 80 mg/L, after 5 hours = 40 mg/L, after 10 hours = 20 mg/L

Calculate how much of the drug remains in the body after 10 hours
Cp = Dose/Vd
Dose = Cp * Vd
Dose = 20mg/L * 13.75 L = 275 mg

38

QUESTION #7

MN is 57-year-old female who weighs 145 pounds. She was admitted to the hospital with a severe UTI and was given an IV dose of cephalexin 500 mg. Two hours later, a blood sample was withdrawn to measure the drug concentration. If all the medication was present in the body and not eliminated during the 2 hours, what is the expected peak plasma concentration following the single cephalexin dose? Assume the volume of distribution to be 0.2 L/kg and the salt factor to be 1.

39

QUESTION #7 - ANSWER

MN is 57-year-old female who weighs 145 pounds. She was admitted to the hospital with a severe UTI and was given an IV dose of cephalexin 500 mg. Two hours later, a blood sample was withdrawn to measure the drug concentration. If all the medication was present in the body and not eliminated during the 2 hours, what is the expected peak plasma concentration following the single cephalexin dose? Assume the volume of distribution to be 0.2 L/kg and the salt factor to be 1.

Convert her weight to kg: 145/2.2 = 65.9 kg

Vd = 0.2 L/kg (65.9kg) = 13.18 L

For IV dose, C(peak)=Cp0 = Dose/Vd = 500 mg / 13.18L = 37.93 mg/L

40

QUESTION #8

EP is a 68 kg patient who is started on an IV infusion of a new antibiotic. If the target plasma concentration for this patient is 4 mg/L, calculate the required rate of infusion for EP given that its volume of distribution = 30 L, half–life is 17 h, and clearance is 2 L/h? If you want to achieve the target plasma concentration quickly, what loading dose would you need to give?

41

QUESTION #8 - ANSWER

EP is a 68 kg patient who is started on an IV infusion of a new antibiotic. If the target plasma concentration for this patient is 4 mg/L, calculate the required rate of infusion for EP given that its volume of distribution = 30 L, half–life is 17 h, and clearance is 2 L/h? If you want to achieve the target plasma concentration quickly, what loading dose would you need to give?

Cp,ss = 4 mg/L

Cp,ss = R/Cl

R = Cp,ss * Cl = 4 mg/L * 2 L/h = 8 mg/h

LD = Cp,ss * Vd = 4 mg/L * 30 L = 120 mg

42

QUESTION #9

PA was infused with a medication (Med X) for a duration of 6 hours at a rate of 2 mg/hr. The medication has a volume of distribution (Vd) of 10 L and an elimination rate constant (k) of 0.02 hr–1. What is the concentration of Med X in PA's body 2 hours after the infusion was stopped?

43

QUESTION #9 - ANSWER

PA was infused with a medication (Med X) for a duration of 6 hours at a rate of 2 mg/hr. The medication has a volume of distribution (Vd) of 10 L and an elimination rate constant (k) of 0.02 hr–1. What is the concentration of Med X in PA's body 2 hours after the infusion was stopped?

Equation for IV infusion: $\mathbf{Cp} = \frac{\mathbf{R_0}}{\mathbf{Cl}} \cdot (\mathbf{1} - \mathbf{e^{-kt}}) * \mathbf{e^{-kt}}$

Cl = kVd = 0.02*10 = 0.2 L/hr

$\mathbf{Cp} = \frac{2\ mg/hr}{0.2\ L/hr} \cdot (\mathbf{1} - \mathbf{e^{-0.02(6)}}) * \mathbf{e^{-0.02(2)}}$

Cp= 1.1 mg/L

44

QUESTION #10

SM is a 45-year-old male patient with a history of chronic pain who was administered an intravenous (IV) bolus dose of a medication. Ten hours post-administration, the drug's plasma concentration was measured at 4 mg/L. Given a known volume of distribution (Vd) of 10 L and an elimination half-life (t1/2) of 5 hours, calculate the dose administrated to the patient.

45

QUESTION #10 - ANSWER

SM is a 45-year-old male patient with a history of chronic pain who was administered an intravenous (IV) bolus dose of a medication. Ten hours post-administration, the drug's plasma concentration was measured at 4 mg/L. Given a known volume of distribution (Vd) of 10 L and an elimination half-life (t1/2) of 5 hours, calculate the dose administrated to the patient.

Half life is 5 hours, Cp fell to 4 mg/L after 10 hours (2 half lives).

16 mg/L (t=0) → 8 mg/L (after 5 hours) → 4 mg/L (after 10 hours)

Dose = Cp0 * Vd = 16 mg/L * 10L = 160 mg

46

QUESTION #11

A 30-year-old female presents to the emergency department complaining of severe dysmenorrhea. The physician decides to administer a loading dose of an analgesic medication to promptly alleviate her pain. The analgesic administered has the following pharmacokinetic characteristics: a half-life of 10 hours, a clearance rate of 0.04 L/min, and a volume of distribution of 40 L. To achieve complete pain relief, the target plasma concentration for the drug has been set at 4 mcg/mL. Calculate the loading dose needed for this patient to ensure rapid pain relief.

QUESTION #11 - ANSWER

A 30-year-old female presents to the emergency department complaining of severe dysmenorrhea. The physician decides to administer a loading dose of an analgesic medication to promptly alleviate her pain. The analgesic administered has the following pharmacokinetic characteristics: a half-life of 10 hours, a clearance rate of 0.04 L/min, and a volume of distribution of 40 L. To achieve complete pain relief, the target plasma concentration for the drug has been set at 4 mcg/mL. Calculate the loading dose needed for this patient to ensure rapid pain relief.

Cp = 4 mcg/mL

Dose = Cp * Vd

Vd=40 * 1000mL = 40000mL

Dose = 4 mcg/mL * 40000 mL = 160000 mcg = 160 mg

QUESTION #12

What is the necessary infusion rate for aminophylline considering that the desired steady-state plasma concentration is 15 mg/L? The average half-life is about 4 h and the apparent volume of distribution is about 25 liters.

49

QUESTION #12 - ANSWER

What is the necessary infusion rate for aminophylline considering that the desired steady-state plasma concentration is 15 mg/L? The average half-life is about 4 h and the apparent volume of distribution is about 25 liters.

$$Cp = \frac{R_0}{Cl} \cdot \left(1 - e^{-kt}\right)$$

Cp,ss= 15 mg/L

Cl = kVd

K = ln2/$t_{1/2}$ = ln2/4 = 0.173 h-1

Cl= 0.173 * 25 = 4.33 L/h

Cp,ss * Cl = R0

15 * 4.33 ~ 65 mg/h

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50

QUESTION #13

MJ has been suffering from a severe infection in his right foot. The doctor decided to give him a single dose of 220 mg of an antibiotic in the form of an infusion that lasts 60 min. The plasma concentration after one hour of the infusion is 14.4 mg/L. How many hours after the end of infusion will the concentration reach 2mg/L, considering that the Kel=0.35 hr-1? What is the half life of this antibiotic?

Cp0=14.4 mg/L

Cp(t) = 2 mg/L

t = ?

51

QUESTION #13 - ANSWER

MJ has been suffering from a severe infection in his right foot. The doctor decided to give him a single dose of 220 mg of an antibiotic in the form of an infusion that lasts 60 min. The plasma concentration after one hour of the infusion is 14.4 mg/L. How many hours after the end of infusion will the concentration reach 2mg/L, considering that the Kel=0.35 hr-1? What is the half life of this antibiotic?

Cp (t) = $Cp_0 * e^{-kt}$

2 = 14.4 * $e^{-(0.35)t}$

T = ln (2/14.4) / -0.35

T = 5.71 h to reach 2 mg/L

$T_{1/2}$ = ln(2)/0.35 = 1.98 h

52

QUESTION #14

In a phase II clinical trial, a 20-mg dose of a newly developed medication exhibiting first-order elimination kinetics was administered intravenously to healthy subjects. The drug's volume of distribution (Vd) was determined to be 50 liters. If a 40-mg dose of the same medication is administered, what is the calculated volume of distribution for this higher dose?

a) 100 L
b) 50 L
c) 25 L
d) 125 L

53

QUESTION #14 - ANSWER

In a phase II clinical trial, a 20-mg dose of a newly developed medication exhibiting first-order elimination kinetics was administered intravenously to healthy subjects. The drug's volume of distribution (Vd) was determined to be 50 liters. If a 40-mg dose of the same medication is administered, what is the calculated volume of distribution for this higher dose?

Vd is CONSTANT for a drug (i.e. does not change with the dose) = 50 litres for a 40 mg dose as well

54

QUESTION #15

JG, a 60-year-old patient who weighs 75 kg, is receiving an intermittent IV infusion of Drug Y at a dose of 300 mg. After the administration of Drug Y, JG's peak plasma concentration is measured and found to be 26 mg/L. The elimination rate constant for Drug Y is determined to be 0.2 h-1 and volume of distribution is 0.8 L/kg. Calculate the total clearance (mL/min) of Drug Y in JG.

55

QUESTION #15 - ANSWER

JG, a 60-year-old patient who weighs 75 kg, is receiving an intermittent IV infusion of Drug Y at a dose of 300 mg. After the administration of Drug Y, John's peak plasma concentration is measured and found to be 26 mg/L. The elimination rate constant for Drug Y is determined to be 0.2 h-1 and volume of distribution is 0.8 L/kg. Calculate the total clearance (mL/min) of Drug Y in JG.

Cl = kVd = 0.2 h-1 * 0.8 L/kg (75kg) = 12 L/h

12 L/h *(1000mL/L)* (1h/60min) = 200 mL/min

56

QUESTION #16

A patient received a new antibiotic at a dosage of 10 mg/kg twice daily, at 09:00 and 21:00, administered as a 45-minute intravenous infusion. The peak concentration was evaluated at 11:00 and measured as 40 mcg/dL. The trough level was measured 1 hour before the subsequent dose, and it was observed to be 3 mcg/dL. What is the drug's half-life (t½)?

57

QUESTION #16 - ANSWER

A patient received a new antibiotic at a dosage of 10 mg/kg twice daily, at 09:00 and 21:00, administered as a 45-minute intravenous infusion. The peak concentration was evaluated at 11:00 and measured as 40 mcg/dL. The trough level was measured 1 hour before the subsequent dose, and it was observed to be 3 mcg/dL. What is the drug's half-life (t½)?

$$\frac{Cp_{max,ss}}{Cp_{min,ss}} = \frac{1}{e^{-k\tau}}$$

$$\frac{40}{3} = \frac{1}{e^{-k(12)}}$$

K = ln(3/40)/-12 = 0.216 h-1

$T_{1/2}$= ln2/0.216 = 3.21 h

58

QUESTION #17

JR is a 32-year-old female who weighs 150 pounds needs an intravenous infusion of antibiotic (X). The physician would like to achieve a steady state concentration (Css) of 15 mg/dl by infusing the dose over 8 hours without giving a loading dose. Drug (X) has an elimination t½ of 1.4 hours and a volume of distribution (VD) of 8 L.

A) What is the rate of intravenous infusion that should be recommended for this patient?

B) How long after starting the IV infusion would the plasma concentration reaches 95% of the desired steady state concentration (Css)?

C) What is the recommended loading dose?

D) What is the total clearance rate of this antibiotic in this patient in mL/min?

E) If the patient's renal clearance of this antibiotic is 40 mL/min, what is the hepatic clearance?

59

QUESTION #17 - ANSWERS

JR is a 32-year-old female who weighs 150 pounds needs an intravenous infusion of antibiotic (X). The physician would like to achieve a steady state concentration (Css) of 15 mg/dl by infusing the dose over 8 hours without giving a loading dose. Drug (X) has an elimination t½ of 1.4 hours and a volume of distribution (VD) of 8 L.

A) What is the rate of intravenous infusion that should be recommended for this patient?

$Cp = \frac{R_0}{Cl}$, R_0= Cp*Cl = 150 mg/L * 3.96 L/h = 5.94 mg/h

Cl = kVd = ln2/1.4h * 8 = 3.96 L/h

B) How long after starting the IV infusion would the plasma concentration reaches 95% of the desired steady state concentration (Css)?

5 * 1.4h = 7 hours

C) What is the recommended loading dose?

150 mg/L * 8 = 1200 mg

60

QUESTION #17 - ANSWERS

JR is a 32-year-old female who weighs 150 pounds needs an intravenous infusion of antibiotic (X). The physician would like to achieve a steady state concentration (Css) of 15 mg/dl by infusing the dose over 8 hours without giving a loading dose. Drug (X) has an elimination t½ of 1.4 hours and a volume of distribution (VD) of 8 L.

D) What is the total clearance rate of this antibiotic in this patient in mL/min?

3.96 L/h or 66 mL/min

E) If the patient's renal clearance of this antibiotic is 40 mL/min, what is the hepatic clearance?

26 mL/min

QUESTION #18

SJ, a 30-year-old female, was administered a dose of Drug (Y), and two hours later, her serum drug concentration was found to be 43.2 mg/L. Ten hours after the dose was given, her serum drug concentration was measured again and found to be 12 mg/L. Assuming first-order kinetics for the drug, what is the half-life of drug (Y)?

QUESTION #18

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k = ln(43.2mg/l)-ln(12) / 10 h = 0.128 h-1

T1/2 = ln2/k = ln2 / 0.128h-1 = 5.42h

63

QUESTION #19

In a clinical setting, a male patient who is 50 years of age was administered a single intravenous dose of 10 mg of an antibiotic. The serum drug concentration was subsequently assessed and found to be 75 mcg/dL. Given a patient weight of 180 pounds, what is the volume of distribution (Vd) in L/kg of the administered antibiotic?

64

QUESTION #19 - ANSWER

In a clinical setting, a male patient who is 50 years of age was administered a single intravenous dose of 10 mg of an antibiotic. The serum drug concentration was subsequently assessed and found to be 75 mcg/dL. Given a patient weight of 180 pounds, what is the volume of distribution (Vd) in L/kg of the administered antibiotic?

Weight 180 lbs → 81.81 kg

Dose 10 mg → 10,000 mcg

Concentration 75 mcg/dL → 750 mcg/L

Vd = Dose/conc = 10000/750 = 13.33 L / 81.81 kg = 0.163 L/kg

QUESTION #20

After administration of an antibiotic to patient KS, the plasma drug concentration was measured at 4 μg/mL. If the clearance of the antibiotic is 20 mL/min, what is the elimination rate for the antibiotic?

QUESTION #20 - ANSWER

After administration of an antibiotic to patient KS, the plasma drug concentration was measured at 4 µg/mL. If the clearance of the antibiotic is 20 mL/min, what is the elimination rate for the antibiotic?

Rate of elimination = CL * Cp = 20 ml/min * 4 ug/mL = 80 ug/min

67

QUESTION #21

Tobramycin 240 mg injection given by IV and plasma concentrations were determined as follows:

Time After Injection (hour)	Concentration (mg/litre)
1	21.6
2	17.8
3	14.8

What is the initial plasma concentration of the drug?

68

QUESTION #21 - ANSWER

Tobramycin 240 mg injection given by IV and plasma concentrations were determined as follows:

Time After Injection (hour)	Concentration (mg/litre)
1	21.6
2	17.8
3	14.8

What is the initial plasma concentration of the drug?

Plot on ln graph → yield straight line → take slope (k) = ln (14.8)-ln (21.6) / (3-1) = 0.189 h-1

Cp (t) = $Cp_0 * e^{-kt}$

Cp0 = Cp(t)/e^-kt (plug in any data point)

Cp0 = 21.6/e^-(0.189)(1)

Cp0 = 26 mg/L

69

QUESTION #22

CY is a 29-year-old female (165cm height, 65 kg weight), she was diagnosed with bladder cancer and is started on the following chemo regimen:

- Cycle: 28 days – repeat for 5 cycles
- Methotrexate – 30 mg/m^2 on days 1, 15 ,22
- Vinblastine, 3mg/m^2 on days 1, 15, 22
- Doxorubicin, 30mg/m^2 on day 2
- Cisplatin, 70mg/m^2 on day 2

a) Calculate the total amount of each medication that is administered per cycle

b) What is the IV infusion rate if the total amount of doxorubicin is delivered in 4 hours from a 1L solution

c) Calculate the steady state concentration of doxorubicin, assuming it is 100% renally cleared

d) The pKa of methotrexate is 4.8. The ratio of ionized to unionized methotrexate in the IV solution is 75:1. What is the pH of this solution?

e) If an injection of cisplatin containing 8 mg/mL is available, what is the volume of injection required to treat the patient?

70

QUESTION #23

The oral dose of a drug is 3 mg. The solution contains 0.5% w/v of the drug in a dropper bottle that delivers 15 drops/mL, with a total volume of 30 mL. The drug is taken BID for 30 days. How many bottles needed to dispense?

71

QUESTION #24

A concentrated acid consists of 60% w/w acid and has a specific gravity of 1.56.

a) How many milliliters of the acid would be required to prepare 55 mL of a 10% w/v solution?
b) If the strength of the acid is mistakenly read as 60% w/v, how much of the concentrated acid would be used to prepare the solution in (a)?
c) What would be the percent error in the amount of concentrated glycolic acid measured in b)
d) What would be the resulting percent strength of the diluted acid solution in b) due to the mistake?

72

PHARMACOKINETIC EQUATIONS

Equation	Comments
$C_t = C_0 \cdot e^{-k_{el} \cdot t}$	C_t = concentration at time t (mg/L) C_0 = concentration at the end of the infusion (mg/L) t = any time after the end of the infusion (hours)
$C = C_0 * 10^{-\frac{kt}{2.303}}$	C= concentration ; C_0 = initial concentration k= rate constant T= Time at concentration "C"
$C_{min} = C_t \cdot e^{-k_{el} \cdot t_2}$	C_{min} = extrapolated trough concentration (mg/L) C_t = measured concentration (mg/L) t_2 = time from measured trough to start of next dose (h)
$logC = logC_0 - \frac{k}{2.303}t$	C= concentration ; C_0 = initial concentration k= rate constant T= Time at concentration "C"
$1 - e^{-k_{el} \cdot t}$	Fraction lost during decay or fraction achieved during the infusion
$C_{max} = C_p \cdot e^{-k_{el} \cdot t_1}$	C_{max} = extrapolated peak concentration (mg/L) C_p = measured peak concentration (mg/L) t_1 = time from end of infusion to measured peak (h)

73

PHARMACOKINETIC EQUATIONS

Equation	Comments
$V_{ss} = \left(\frac{Dose}{k_{el}\, t_i}\right)\left(\frac{(1 - e^{-kt_i})}{C_{max} - (c_{min} \times e^{-kt_i})}\right)$	V_{ss} = volume of distribution at steady-state (L) Dose = administered dose (mg) t_i = infusion time (hours) k_e = elimination rate constant (h^{-1}) C_{max} = extrapolated peak concentration (mg/L) C_{min} = extrapolated trough concentration (mg/L)
$t_{1/2} = \frac{0.693 \times V_{ss}}{Cl_{ss}}$	$t_{½}$ = half-life (hours) Cl_{ss} = clearance of drug in steady-state (L/h) V_{ss} = volume of distribution at steady-state (L)
$\text{New } C_{tss} = \text{New Cpss} \cdot e^{-k_{el} \cdot T}$	New C_{tss}=trough concentration at SS (mg/L) k_{el} = elimination rate constant (h^{-1}) $T = n\tau - (t_i + t_1)$ (h)

74

PHARMACOKINETIC EQUATIONS

Equation	Comments
$Cl_{ss} = k_{el} \cdot V_{ss}$	Cl_{ss} = clearance of drug in steady-state (L/h) k_{el} = elimination rate constant (h^{-1}) V_{ss} = volume of distribution at steady-state (L)
$k_{el} = Cl_{ss}/V_{ss}$	Cl_{ss} = clearance of drug in steady-state (L/h) K_{el} = elimination rate constant (h^{-1}) V_{ss} = volume of distribution at steady-state (L)
$E_\tau = \frac{\ln(C_{pd}/C_{td})}{k_{el}} + (t_i + t_1)$	E_τ = estimated new dosing interval (hours) k_e = elimination rate constant (h^{-1}) C_{pd} = desired peak concentration (mg/L) C_{td} = desired trough concentration (mg/L) t_i = infusion time (hours) t_1 = time from the end of infusion to measure peak (hours)

75

PHARMACOKINETIC EQUATIONS

Equation	Comments
$\text{New Dose} = ti \times CLss \times \frac{C_{pd}}{e^{-kt_1}} \times \frac{1 - e^{-kn\tau}}{1 - e^{-kti}}$	New Dose = estimated new dose (mg) T_i = infusion time (hours) K_e = elimination rate constant (h^{-1}) CL_{ss}= clearance of drug at steady-state (L/h) C_{pd} = desired peak concentration (mg/L) t_1 = time from the end of infusion to measured peak (hours) $n\tau$ = new dosing intervals (hours)
$A = \text{Dose} \cdot e^{-k_{el} \cdot t}$	A = amount of drug remaining in the body Dose = dose k_{el} = elimination rate constant t = time $t_{1/2}$ = elimination half-life

76

PHARMACOKINETIC EQUATIONS

Equation	Comments
$C_{ss} = \frac{R_0}{CL}$	C_{ss} = blood concentration at steady-state R_0 = the infusion rate CL = whole-body clearance
$A_{ss} = \frac{R_0}{k_{el}}$	A_{ss} = amount of drug in the body at SS k_{el} = the elimination rate constant
$\text{New } Cp_{ss} = \left(\frac{\text{New dose}}{t_i} + CL_{ss}\right)\left(\frac{1 - e^{-kti}}{1 - e^{-kn\tau}}\right) \cdot e^{-k_{el} \cdot t_1}$	New Cp_{ss} = predicted peak concentration at steady-state (mg/L) New Dose = chosen new dose (mg) t_i = infusion time (hours) k_{el} = elimination rate constant (h^{-1}) CL_{ss} = clearance of drug at steady-state (L/h) t_1 = time from the end of infusion to measured peak (hours) Nτ = new dosing intervals (hours)

77

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78

CHANGE LOG

September 2024 – Review Session

- Slide 3: Updated legend
- Added IV bolus, multiple dose slides
- Added non-linear PK slides as not covered in PK2
- New questions with answers on s21-33
- S34-68 – pulled PK questions from bonus practice problems and created answer slides for each question
- S69-71 – extra calculations questions pulled from PA handbook

October 2025

- Slide 18: Used Actual body weight instead of ideal, corrected answer to 1003 mg
- Slide 38 : Fixed solution for Part C

79